FORMAL ETALENESS IS EQUIVALENT TO THE VANISHING OF BOTH COHOMOLOGY SHEAVES OF THE NAIVE COTANGENT COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5471-5478 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 5480-5492, the complete author proof (11 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The naive cotangent complex, its cohomology sheaves, and the characterisations of formal smoothness and formal unramifiedness by them; the combination is retained.

Source reference. Original lemma lemma-NL-formally-smooth: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5451–5459. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-characterize-formally-unramified: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2810–2816. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-formally-etale-unramified-smooth: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4539–4543. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. The following are equivalent*

(1) *f is formally étale,*
(2) $H^{-1}(NL_{X/Y}) = H^0(NL_{X/Y}) = 0$.

Proof. Assume (1). A formally étale morphism is a formally smooth morphism. Thus $H^{-1}(NL_{X/Y}) = 0$ by Lemma lemma-NL-formally-smooth. On the other hand, a formally étale morphism if formally unramified hence we have $\Omega_{X/Y} = 0$ by Lemma lemma-characterize-formally-unramified. Conversely, if (2) holds, then

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f is formally smooth by Lemma lemma-NL-formally-smooth and formally unramified by Lemma lemma-characterize-formally-unramified and hence formally étale by Lemmas lemma-formally-etale-unramified-smooth. $\square$